Mathski

IA Self-Check

Mathematics: analysis and approaches · applications and interpretation · SL and HL

Use this before you hand in a draft. Tick each statement only if you can point to it in your IA, then estimate your mark for each criterion. IAs are marked holistically against the IB descriptors, so one unticked box does not cap your mark, and ticking every box does not guarantee the top band.

A Presentation

My estimate ____ / 4

WHAT THE MARKS LOOK LIKE

1. Some structure or order, but hard to follow.
2. Some structure and some order, with gaps.
3. Clear and well organised throughout.
4. Clear, well organised **and concise**: every section earns its place.

CAN I POINT TO THIS IN MY IA?

- ☐ My introduction states a specific aim (what I will determine, compare or model), not just a topic.
- ☐ I explain why I chose the topic.
- ☐ Headings split the work into sections that follow a logical order.
- ☐ Each step leads on from the last; a classmate on my course could follow it without asking me.
- ☐ Tables, graphs and diagrams sit next to the text that uses them, and each one is explained.
- ☐ Repeated calculations are shown once, with the other results summarised.
- ☐ Nothing is there to fill space: no long history, no results I never use.
- ☐ My conclusion answers the aim from my introduction.
- ☐ Sources are referenced and pages are numbered.

Marker's tip. Length does not earn A marks. Bullet points in place of written explanation usually hold A at 2.

B Mathematical communication

My estimate ____ / 4

WHAT THE MARKS LOOK LIKE

1. Some relevant mathematical communication, partly appropriate.
2. Relevant and appropriate in places.
3. Relevant, appropriate and mostly consistent.
4. Relevant, appropriate and **consistent throughout**.

CAN I POINT TO THIS IN MY IA?

- ☐ I use correct mathematical terms (for example "gradient", not "steepness").
- ☐ Notation is typed properly (x^2, $\sqrt{}$, π), never calculator syntax such as x^2 or sqrt.
- ☐ Every variable is defined, with units, before it is first used.
- ☐ I use more than one form of representation (equations, tables, graphs, diagrams) where it helps.
- ☐ Every graph has a title, labelled axes with units and a sensible scale.
- ☐ Every table has headings and units.
- ☐ I use ≈ for rounded values and give answers to a sensible degree of accuracy.
- ☐ Output from a GDC, Desmos or a spreadsheet is explained, not just pasted in.

Marker's tip. B4 needs consistency all the way through. One unlabelled graph or one undefined variable can hold it at B3.

C Personal engagement

My estimate ____ / 3

WHAT THE MARKS LOOK LIKE

1. Some evidence of personal engagement.
2. Significant evidence.
3. **Outstanding** evidence.

CAN I POINT TO THIS IN MY IA?

- ☐ I explain my personal link to the topic.
- ☐ My interest shapes the mathematics I choose, not just my introduction.
- ☐ I collected or generated my own data, or built my own model, where it makes sense.
- ☐ I ask my own questions, test conjectures and follow up anything surprising.
- ☐ I present ideas in my own way rather than following a textbook or online example.
- ☐ I return to my personal context when I interpret my results.
- ☐ I went beyond the obvious approach: taught myself a method, tried an alternative or extended my aim.

Marker's tip. A personal story in the introduction that disappears afterwards is decorative. However genuine it is, it usually scores C1.

D Reflection

My estimate ____ / 3

WHAT THE MARKS LOOK LIKE

1 Limited reflection.

2 Meaningful reflection.

3 Substantial, **critical** reflection.

CAN I POINT TO THIS IN MY IA?

- ☐ I comment on what each main result means in context.
- ☐ I link my reflection back to my aim.
- ☐ I discuss the limitations of my data, model or method.
- ☐ I explain why I chose each method, and compare it with an alternative where I can.
- ☐ I check that results are reasonable (an estimate, a second method or technology).
- ☐ I suggest specific improvements or extensions, not general ones.
- ☐ Reflection appears throughout the exploration, not only in the conclusion.

Marker's tip. Describing what your results show is D1. Critical reflection that runs through the whole exploration is needed for D3.

E Use of mathematics

My estimate ____ / 6

WHAT THE MARKS LOOK LIKE

1 Some relevant mathematics; limited understanding.

2 Some relevant mathematics, partly correct; some understanding.

3 Mathematics at the level of the course, partly correct; some understanding.

4 Mathematics at the level of the course, **correct**; good understanding.

5 Correct, with the sophistication expected. SL: good understanding. HL: sophistication *or* rigour, thorough understanding.

6 SL: correct, with sophistication and rigour; thorough understanding. HL: **precise**, with sophistication *and* rigour.

CAN I POINT TO THIS IN MY IA?

- ☐ The mathematics is relevant to my aim and at the level of my course (SL or HL).
- ☐ Every calculation and derivation is correct, and I have rechecked any result that later work depends on.
- ☐ I explain why each method is appropriate, not just how I applied it.
- ☐ I interpret my results rather than just stating them.
- ☐ Where I use technology, I show enough working to make clear I understand what it does.
- ☐ Proofs and derivations are set out logically, one step at a time.
- ☐ My work shows rigour: precise definitions, justified assumptions and general results where possible.

Marker's tip. One early error that carries through later work costs heavily: markers credit only the mathematics that is correct.

My estimate

A	B	C	D	E	Total
___ / 4	___ / 4	___ / 3	___ / 3	___ / 6	___ / 20

Next steps. For each criterion where you are below the top band, write one change you will make before the next draft. For a criterion-by-criterion mark and report, upload your IA at **mathski.io**.

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